

- 2 (a) Force is the rate of change of momentum B1
- (b) (i) Resultant force = change of momentum / time taken = $[0.140 \times (5.5 - (-4.0))] / 0.04$ M1
 $= 33 \text{ N}$ A1
- (ii) resultant force on ball = $\Delta p / \Delta t = 33.25 \text{ N}$
- Taking forces on the ball, (N is force on ball by bar)*
- $33.25 = N - W$ M1
- $N = 33.25 + 0.14 \times 9.81 = 34.62 \text{ N}$
- By N3L, force on bar by ball is 35 N. A0
- (c) (i) Taking pivot about support B,
 clockwise moments = $F_A \times (45 - 25)$ C1
- anti clockwise moments = $(0.450 \times 9.81) (25) + 35 (25 + 50)$ C1
- Clockwise moments = anti clockwise moments
 $F_A = 136.768 \text{ N} = 140 \text{ N (to 2 sf)}$ A1
- (ii) net force = 0
- Upward force = downward force
- $F_B = 35 + 140 + 0.450 \times 9.81$ M1
- $= 180 \text{ N (to 2 sf)}$ A1

3 (a) $\text{Pressure} = \rho gh = 900 \times 9.81 \times 6.0/100$ M1